MODULE 24 · DOMAIN 1 - AGENTS AND WORKFLOWS

Orchestrators and Subagents

Lead agents and subagents, manager and supervisor hierarchies, isolated context, the subagent brief, handoff contracts, budgets, the token cost of multi-agent systems, and when one agent is better.

DOMAIN 1 of 8 EXAM WEIGHT 14.7% MODULE 24 of 36

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A head chef calls orders to stations. Each station cooks one thing well, and sends back a plate, not its whole kitchen.

Orchestrator and subagents

01
Plan
The lead agent breaks the task into parts.

02
Brief
Each subagent gets a clear, bounded task.

03
Work
Subagents run in their own contexts, often in parallel.

04
Return
Each returns a condensed result, checked against a contract.

05
Synthesise
The lead combines, checks gaps, and decides what next.

ON THE EXAM

Anthropic: "a lead agent coordinates the process while delegating to specialized subagents that operate in parallel".

Why isolated context matters

- **Each subagent starts clean:** no parent history, no parent system prompt, just its brief.
- **Only its final message returns,** so hundreds of tool results never reach the lead.
- **Summaries are small:** often one to two thousand tokens.
- **Large outputs go to storage;** subagents pass back lightweight references.
- **The flip side:** a subagent knows only what the brief tells it.

> THE TIEBREAKER
>
> Isolation keeps the lead's context lean and focused. It also means the brief must carry everything needed.

The subagent brief

- **An objective:** exactly what to find out or produce.
- **An output format:** the shape of what comes back, ideally a schema.
- **Guidance on tools and sources:** where to look, and where not to.
- **Clear task boundaries:** what is in scope, and what another subagent covers.
- **Vague briefs cause duplicated work:** three subagents researching the same thing.

> ON THE EXAM
>
> Anthropic: "Each subagent needs an objective, an output format, guidance on the tools and sources to use, and clear task boundaries."

A handoff contract

```python
class PaymentsFindings(BaseModel):                  # what the payments subagent returns
    tenancy_id: str
    arrears_pence: int                              # computed in code, not by the model
    missed_payments: list[MissedPayment]
    sources: list[str]                              # record IDs, so the lead can check
    open_questions: list[str]                       # what it could not settle
    status: Literal["complete", "partial", "failed"]

findings = PaymentsFindings.model_validate_json(subagent_result)   # gate
if findings.status != "complete": lead_notes.append(findings.open_questions)
```

5-7 Sources, gaps and a status, not just a confident paragraph.

9-10 Validated before the lead uses it; partial results handled.

THE TAKEAWAY

A schema between agents is the same contract as between workflow stages. Validate every handoff.

Manager or supervisor?

MANAGER HIERARCHY

The lead delegates

- A lead plans and assigns work
- Workers report back to it
- Can nest: managers of managers
- Each level adds cost and latency

SUPERVISOR

One agent checks another

- A worker acts; a supervisor reviews
- Approves, rejects or escalates
- Often a different model or prompt
- People stay in the loop for risky steps

THE TIEBREAKER

Keep hierarchies shallow. Every extra layer multiplies tokens and hides errors from the top.

The cost of more agents

- **Agents use about 4x the tokens of chat;** multi-agent systems about 15x.
- **The gain can be real:** Anthropic's multi-agent setup beat a single agent by 90.2% on its research eval.
- **Scale effort to the task:** simple fact-finding, 1 agent and 3 to 10 tool calls; complex research, 10 or more subagents.
- **Budgets per subagent:** turns, tokens, tools, time.
- **Cheaper models for workers;** a stronger model for the lead, if evals agree.

> ON THE EXAM
>
> Multi-agent pays off when the task is valuable, broad and parallel. Otherwise, 15x is just 15x.

One agent or several?

SIGNAL	LEAN TOWARDS
Broad task that splits into independent parts	Several agents, in parallel
Too much material for one context	Subagents that return summaries
Steps depend tightly on each other	One agent
Every step needs the same shared context	One agent
Low value per task, high volume	One agent, or a workflow

THE TIEBREAKER

Anthropic: tasks where agents must "share the same context" or have "many dependencies" fit multi-agent poorly. Most coding is one.

Copperlet's agency-ops agent, with subagents

THE LEAD

Opus 5.5, plans and writes

- Reads the case, plans three lines of enquiry
- Briefs each subagent with objective, format, sources, scope
- Validates each findings schema
- Writes the summary for the letting agent

THREE SUBAGENTS

Sonnet 5.5, read-only

- Payments: ledger and missed payments
- Messages: tenant and agent history
- CRM and email: promises and disputes
- Each: 15 turns, token budget, findings schema

THE TAKEAWAY

Big cases fan out; a simple case with one payment missed stays with the lead alone.

> A lead research agent gives each of four subagents the brief "research our competitors". The subagents return overlapping reports, two cover the same company, and none use the format the lead expects.
>
> The team wants better results without more agents.

What should change first?

A Give each a specific objective, format, sources and boundaries

B Add a fifth subagent to remove duplicates afterwards

C Move all four subagents to the most capable model

D Share the lead's full conversation history with every subagent

◆ PAUSE AND DECIDE

What should change first?

A Give each a specific objective, format, sources and boundaries

Fixes overlap and format at the source.

B Add a fifth subagent to remove duplicates afterwards

More agents, same vague briefs.

C Move all four subagents to the most capable model

A better model, still told the same thing.

D Share the lead's full conversation history with every subagent

Loses isolation; still no boundaries.

THE SIGNAL

Overlap and wrong formats come from the brief. Anthropic's four ingredients fix both.

A single agent investigating insurance claims reads policy documents, call transcripts and payment logs. By the middle of each case its context is full of raw tool output, and its final reasoning gets worse.

The three sources can be examined independently.

What design helps most?

A. Raise max_tokens so the agent can write longer reasoning

B. Paste all three sources into the first message

C. A subagent per source, each returning a checked summary

D. Run the same single agent three times and compare

◆ PAUSE AND DECIDE

What design helps most?

A Raise max_tokens so the agent can write longer reasoning

Output limit; input is the problem.

B Paste all three sources into the first message

Even more bloat, sooner.

C A subagent per source, each returning a checked summary

Isolation keeps the lead lean.

D Run the same single agent three times and compare

Three bloated contexts.

THE SIGNAL

Context full of raw output, sources independent: subagents that return summaries.

A team proposes a five-agent system for answering simple internal FAQ questions, such as office opening hours. Each question would go through a planner, two researchers, a writer and a reviewer.

Volume is high and answers are short.

What should the team do?

A Build the five-agent system, since more agents means better answers

B Use a single call, or one agent with retrieval, and check it against the FAQ evals

C Use the five agents, but on the most capable model

D Add a sixth agent to supervise the other five

◆ PAUSE AND DECIDE

What should the team do?

A Build the five-agent system, since more agents means better answers

About 15x the tokens for a lookup.

B Use a single call, or one agent with retrieval, and check it against the FAQ evals

Effort scaled to the task.

C Use the five agents, but on the most capable model

Multiplies an already high cost.

D Add a sixth agent to supervise the other five

Even more layers for a simple answer.

THE SIGNAL

Simple, high-volume, low-value: one call or one agent. Scale effort to complexity.

> In a multi-agent pipeline, a subagent sometimes returns confident prose with no sources and a made-up total. The lead copies it into the final report.
>
> The team wants bad handoffs caught before they spread.

What should the team add?

A. Validate a results schema with sources; compute totals in code

B. An instruction asking the subagent to be more careful

C. A longer context window for the lead

D. More subagents covering the same task, keeping the most confident answer

◆ PAUSE AND DECIDE

What should the team add?

A Validate a results schema with sources; compute totals in code

A checked contract; exact work in code.

B An instruction asking the subagent to be more careful

Guidance, not a gate.

C A longer context window for the lead

Room is not the problem.

D More subagents covering the same task, keeping the most confident answer

Confidence is not accuracy.

THE SIGNAL

Handoffs between agents need the same gates as handoffs between stages.

> A team is choosing between one agent and an orchestrator with subagents for two tasks. Task one is a coding change where every edit depends on the previous one and all steps need the same files. Task two is a market scan across twenty independent regions.
>
> The team's budgets are moderate.

Which mapping is best?

A. Multi-agent for both, to keep the design consistent

B. One agent for both, to keep costs down

C. Multi-agent for the coding change, one agent for the market scan

D. One agent for the coding change; subagents for the scan

◆ PAUSE AND DECIDE

Which mapping is best?

A Multi-agent for both, to keep the design consistent

Wrong fit for the coding change.

B One agent for both, to keep costs down

Twenty regions overload one context.

C Multi-agent for the coding change, one agent for the market scan

Exactly backwards.

D One agent for the coding change; subagents for the scan

Each task gets its fitting shape.

THE SIGNAL

Dependent steps, shared context: one agent. Broad, independent parts: subagents.

> A multi-agent research tool works well, but costs are three times the forecast. Logs show subagents running up to 80 turns, and all of them use the largest model.
>
> Quality must stay at its current level.

Which two changes fit best? Choose two.

A Remove the lead agent, and let subagents coordinate among themselves

B Set per-subagent budgets for turns, tokens and time

C Add another layer of managers to watch costs

D Share every subagent's full transcript with the lead

E Evaluate a cheaper model for the subagents only

◆ PAUSE AND DECIDE

Which two changes fit best? Choose two.

A Remove the lead agent, and let subagents coordinate among themselves

Loses planning and synthesis.

B Set per-subagent budgets for turns, tokens and time

Caps the runaway turns.

C Add another layer of managers to watch costs

More agents, more cost.

D Share every subagent's full transcript with the lead

Bloats the most expensive context.

E Evaluate a cheaper model for the subagents only

Most tokens at the lower rate.

THE SIGNAL

Two cost signals: runaway turns and the largest model everywhere. Budget the first; re-tier the second.

Module 24 in five lines

- **Lead plans, subagents work in isolated contexts,** results come back condensed.
- **Every brief:** objective, output format, tools and sources, boundaries.
- **Handoffs are contracts:** schemas with sources and status, validated by the lead.
- **About 15x the tokens of chat:** budget each subagent, tier the models.
- **One agent** for dependent, shared-context or simple work; several for broad, parallel work.

COMING UP

Next: the Claude Agent SDK, and hosting agents.